Shrikaran P

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Professional Summary

Highly motivated Computer Science Engineering student with expertise in Java programming, Object-Oriented Design, Data Structures & Algorithms, SQL, and Full-Stack Development. Skilled in developing robust software applications, designing efficient backend systems, and solving complex technical problems. Hands-on experience through research, cloud computing, machine learning, and enterprise-focused projects, including intelligent monitoring systems, secure data processing pipelines, and predictive analytics applications. Strong analytical, communication, and teamwork skills with a passion for software engineering, cloud technologies, and building scalable business solutions.

Technical Skills

Programming Languages: Java, Python, C, C++
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP)
Web Development: HTML, CSS, JavaScript, React.js
Databases & Cloud: SQL, MySQL,AWS, Git, GitHub
AI/ML: Machine Learning, Deep Learning, PyTorch, CNN-LSTM

Educational Qualifications

B.Tech-Computer science engineering(5th semester) May 2023 - May 2027
SRM Institute of science and Technology - Tiruchchirappalli
CGPA - 9.2/10

Professional Credentials & Experience

Foundations of AI/ML – GeeksforGeeks
Gained hands-on experience in data preprocessing, model building, and evaluation techniques

AWS Academy Cloud Foundations – AWS
Learned fundamentals of cloud computing and data storage for scalable data processing

Java Full Stack Development – EduSkill
Built web-based applications with structured data handling and backend integration

Technical Projects

- **Smart Interconnected Water Dam Alert System** — coordinated multi-dam monitoring with GSM/Wi-Fi, predictive flood control, and centralized supervision.**{Link}**
- **Private LLM–OCR ETL Pipeline for Enterprise Data Warehousing** — secure on-premises pipeline for OCR, document extraction, contextual transformation, and structured warehousing.
- **Motor Imagery EEG Classification (BCI)** — preprocessing .gdf EEG signals, STFT spectrogram generation, and CNN-based left/right hand motor imagery classification.**{Link}**
- **Intrusion Detection System using CICIDS2017** — deep learning–based malicious network traffic detection and comparison with rule-based baselines.**{Link}**
- **Smart Academic Timetable Generator** — automated timetable scheduling system that generates conflict-free academic timetables by allocating subjects, faculty, classrooms, and time slots while satisfying institutional constraints and reducing manual scheduling effort.
- **Farmer Market Price Prediction System** — machine learning–based prediction of crop market prices using historical mandi data, seasonal trends, weather conditions, and demand–supply analysis to help farmers decide the best time and market to sell produce. **{Link}**

Research Inrenship(NITT)

- Motor Imagery EEG Classification (BCI) — Worked with BCI Competition IV Dataset 2a and BCI Competition IV Dataset 2b EEG .gdf recordings for motor imagery analysis.
- Preprocessed raw EEG signals, selected motor cortex channels, and generated STFT-based time–frequency spectrogram representations.
- Developed CNN-based models for left/right hand motor imagery classification and subject-wise performance evaluation.

Research & Publications

- A Secure On-Premises ETL Pipeline Using OCR and Local LLMs - ICCMC 2025 **{Link}**
- Artificial Intelligence in Smart Agriculture | Indian Journal of Natural Sciences (Oct 2024)

Languages: English,Tamil